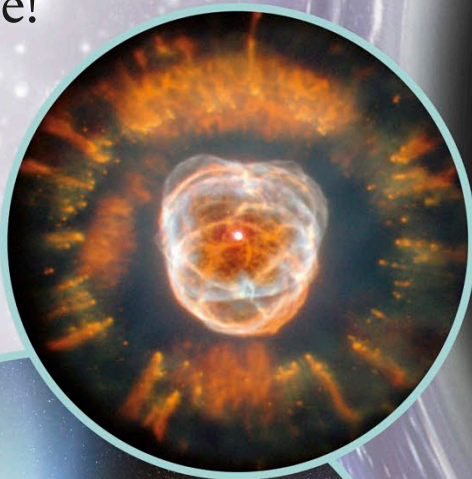


Black holes

Black holes are a great mystery. Astronomers know they are there because of their effect on nearby stars (if a star is too close, it gets pulled towards the hole), but they are very difficult to study. Why? Because they are black, and that makes them invisible!

When Sun-like stars are dying, they push off their outer layers (right). More massive stars can explode and form a black hole.



Birth of a hole

Black holes are sometimes born when a star explodes and dies. When a star with great mass runs out of fuel, it can't stop gravity pulling its gas together, squeezing it tighter and tighter until it forms a tiny neutron star, or a black hole.

Into the hole

Black holes have such strong gravity that nothing in the surrounding space can escape, not even light. However, they do not act as enormous vacuum cleaners – something has to get close enough to be in danger of being pulled in. They are a bit like space whirlpools, affecting just their area of space.



Sometimes, some of the falling gas is squirted back out as hot jets.

This artist's idea of a black hole shows clouds of gas and dust swirling rapidly around it before being pulled in towards the hole at the centre.